Reassessment of the microTEDAclus Clustering Algorithm and Proposal for an Improved Model

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Abstract

The abstract is deferred to after we finish the main content of the document. This allows us to write a more informed and concise summary of the work after all sections have been completed.

1 Introduction

The introduction is also deferred until we have a clearer picture of the entire document. We will revisit this section after we have completed the main body of the report to ensure it effectively sets the stage for our research and findings.

2 Background and Related Work

The TEDA framework is presented as an alternative to probability theory for handling data in non-deterministic scenarios [3]. The original microTEDAclus model applies its metrics to decide if a sample is an outlier to a microcluster and if microclusters are suitable candidates to form a macrocluster [8]. Finally, when evaluating the previous model's performance in the last-mile problem, [6] refactor the decay logic and variance computation.

2.1 Typicality and Eccentricity-Based Data Analytics

The paper *Outside the Box: An Alternative Data Analytics Frame-Work*, by Plamen Angelov, introduces the mathematical formulation for the metrics **Eccentricity** and **Typicality** [3].

Let

$$C = \{\sum_{i=1}^{k} \pi_i^k > 0, \quad j > 1, \quad k > 2\} \tag{1}$$

be a set of conditions for a sample j to be considered for the computation of eccentricity and typicality in a moment k.

Eccentricity, a measure of how much a sample deviates from a cluster, is defined as:

$$\xi_j^k = \frac{2\pi_j^k}{\sum_{i=1}^{k} \pi_i^k}, \quad C \tag{2}$$

Where π_j^k is the sum of distances from the sample j to all samples in the moment k.

Typicality, a measure of how well a sample fits within a cluster, is defined as:

$$\tau_j^k = 1 - \xi_j^k, \quad C \tag{3}$$

Both are, by definition, bounded: $0 < \xi_j^k < 1$ and $0 < \tau_j^k < 1$. The normalized eccentricity and typicality are presented as:

$$\zeta_j^k = \frac{\xi_j^k}{2}, \quad \sum_{i=1}^{k} \zeta_i^k = 1 \quad 0 < \zeta_j^k < \frac{1}{2} \quad C \tag{4}$$

$$t_j^k = \frac{\tau_j^k}{k-2}, \quad \sum_{i=1}^{k} t_i^k = 1 \quad 0 < t_j^k < \frac{1}{k-2} \quad C \tag{5}$$

The paper states that samples for which the normalized eccentricity $\zeta_j^k > 1/k$ are considered anomalous, and those for which the normalized typicality $t_j^k > 1/k$ are considered typical.

Furthermore, it provides recursive, or incremental, formulas for updating $\sum_{i=1}^{k} \pi_i^k$ in the squared Euclidean distance, as follows:

The mean and the scalar product of the samples in the moment k are

$$\mu^k = \frac{k-1}{k}\mu^{k-1} + \frac{1}{k}x_k \tag{6}$$

$$X^k = \frac{k-1}{k}X^{k-1} + \frac{1}{k}\|x_k\|^2 \tag{7}$$

From those,

$$\pi_j^k = k(\|x_j - \mu^k\|^2 + X^k - \|\mu^k\|^2) \tag{8}$$

and

$$\sum_{i=1}^{k} \pi_i^k = \sum_{i=1}^{k-1} \pi_i^{k-1} + 2\pi_k^k \tag{9}$$

Finally, it is made clear those metrics suit various proximity functions, among which are the cosine similarity and the Mahalanobis distance [7].

2.2 Original microTEDAclus

Evolving Clustering Algorithm Based on Mixture of Typicalities for Stream Data Mining, by Maia et al., employs the typicality and eccentricity metrics over the euclidean distance to instantiate and update microclusters from the detection of outliers and typical samples, as well as to decide when to merge microclusters into macroclusters [8].

It redefines the eccentricity expression into a recursive form:

$$\xi(x_k) = \frac{1}{k} + \frac{(\mu_k - x_k)^T (\mu_k - x_k)}{k\sigma_k^2} \tag{10}$$

where

$$\mu_k = \frac{k-1}{k}\mu_{k-1} + \frac{x_k}{k} \tag{11}$$

$$\sigma_k^2 = \frac{k-1}{k}\sigma_{k-1}^2 + \frac{1}{k-1}\|x_k - \mu_k\|^2 \tag{12}$$

Another change is in the criteria regarding the typicality and eccentricity thresholds. **microTEDAclus** considers a sample x_k to be an outlier if

$$\zeta(x_k) > \frac{m^2+1}{2k}, \quad m > 0 \tag{13}$$

In this algorithm, m is an arbitrary function of the number of samples in the data:

$$m(k) = \frac{3}{1 + e^{-0.007(k-100)}} \tag{14}$$

Microclusters are dynamically characterized by the following parameters:

- S_k^i: the number of samples in the microcluster i at moment k.
- μ_k^i: the mean of the samples in the microcluster i at moment k.

- σ_k^2: the variance of the samples in the microcluster i at moment k.
- $\xi^i(x_k)$, $\zeta^i(x_k)$: the eccentricity and normalized eccentricity of the sample x_k to the microcluster i at moment k.
- $\tau^i(x_k)$, $t^i(x_k)$: the typicality and normalized typicality of the sample x_k to the microcluster i at moment k.
- $D_k^i = 1/\zeta^i(x_k)$: the density of the microcluster i at moment k.
- $m_k^i(S_k^i)$: the dynamic threshold function for the microcluster i at moment k.

All the metrics and conditions yet defined for the entire data may be applied to each microcluster. Thus, for $S_k^i > 2$, a sample x_k is considered an outlier to the microcluster i if

$$\zeta^i(x_k) > \frac{m_k^i(S_k^i)^2 + 1}{2S_k^i} \tag{15}$$

and, for $S_k^i = 2$, if

$$\left(\zeta^i(x_k) > \frac{m_k^i(2)^2 + 1}{4}\right) \wedge \left(\left(\sigma_2^i\right)^2 < r_0\right) \tag{16}$$

where r_0 is a hyperparameter of the model.
x_k is checked against all microclusters, which are updated if x_k is not an outlier to them, as follows:

$$\begin{aligned} S_k^i &= S_{k-1}^i + 1 \\ \mu_k^i &= \frac{S_k^i - 1}{S_k^i}\mu_{S_k^i-1}^i + \frac{x_k}{S_k^i} \\ \left(\sigma_k^i\right)^2 &= \frac{S_k^i - 1}{S_k^i}\left(\sigma_{k-1}^i\right)^2 + \frac{1}{S_k^i - 1}\left(\frac{2\|x_k - u_k^i\|}{d}\right)^2 \\ \xi^i(x_k) &= \frac{1}{S_k^i} + \frac{2\left(\mu_k^i - x_k\right)^T\left(\mu_k^i - x_k\right)}{S_k^i\left(\sigma_k^i\right)^2 d} \end{aligned} \tag{17}$$

Macroclusters are the disjoint sets of connected microclusters in a graph where the vertices are the microclusters and the edges are defined by the condition

$$\text{dist}(\mu_k^i, \mu_k^j) < 2\left(\sigma_k^i + \sigma_k^j\right), \quad i \neq j \tag{18}$$

To prevent the collapse of all microclusters into a single macrocluster under overlap, a microcluster m_l^j is a candidate to form an edge only if

$$D_k^l \geq \text{mean}\left(D_k^l\right), \quad l = 1, \ldots, |\mathcal{M}_j| \tag{19}$$

where $\mathcal{M}_j = \left\{m_1^j, m_2^j, \cdots, m_l^j\right\}, \quad l = 1, 2, \cdots, N_j$ is the jth macrocluster composed of a set of N_j connected microclusters m_l^j.
The density of a macrocluster is the sum of the normalized typicalities of its active microclusters:

$$w_k^l = \frac{D_k^l}{\sum_{l \in \mathcal{M}_j} D_k^l} \tag{20}$$

Furthermore, the typicality of a sample x_k to a macrocluster $\mathcal{M}_j$ is the weighted sum of the typicalities of x_k to the microclusters in $\mathcal{M}_j$:

$$\mathcal{T}_j(x_k) = \sum_{l \in \mathcal{M}_j} w_k^l t_k^l(x_k) \tag{21}$$

A sample's predicted label j is the index of the macrocluster $\mathcal{M}_j$ to which it has the highest typicality:

$$j = \arg\max_j \mathcal{T}_j(x_k) \tag{22}$$

No explicit decay logic is defined for microclusters.

2.2.1 Critique

2.2.2 Order-Dependent Density as a Microcluster Characteristic

Because **eccentricity** and **typicality** are metrics between a sample and a microcluster, the density $D_k^i = 1/\zeta^i(x_k)$ is also a metric between a sample and a microcluster. So, it should not be treated as a characteristic of the microcluster itself. Consider a microcluster i at moment k with $\sigma_k^i = \sigma_{\text{small}}$, or for which the radius that contains most of its samples is r_{small}. For any of the standard clustering algorithms, such a microcluster would be considered dense [1, 4, 5], even if the sample x_k is far from its mean μ_k^i. **microTEDAclus**, however, would compute $\zeta^i(x_k)$ as a large value, and therefore D_k^i as a small value, which would make the microcluster i less likely to be activated in the macrocluster formation process. On the other hand, if $\sigma_k^i = \sigma_{\text{large}}$, or if the radius that contains most of its samples is r_{large}, a sample x_k that is close to μ_k^i would have a small $\zeta^i(x_k)$ and therefore a large D_k^i. This would make the microcluster i more likely to be activated, even if it is not dense at all.

2.2.3 Incorrect Condition Logic for Outlier Detection in Microclusters with Two Samples

For $S_k^i = 2$,

$$\mu_2 = \frac{1}{2}(x_1 + x_2) \tag{23}$$

$$\sigma_2^2 = \frac{(x_1 - x_2)^2}{4} \tag{24}$$

$$d(x_2, \mu_2) = 1 \tag{25}$$

$$\zeta(x_2) = 0.5 \tag{26}$$

Because $m \geq 1$,

$$\frac{m_k^i(2)^2 + 1}{4} \geq 0.5 \tag{27}$$

Thus, the condition $\zeta^i(x_k) > \frac{m_k^i(2)^2+1}{4}$ is never satisfied for $S_k^i = 2$, which ensures that any conjunction with it is also never satisfied, and no sample can be considered an outlier to a microcluster with two samples. The simple change of the condition logic from a conjunction to a disjunction would simplify the expression to

$$\left(\sigma_2^i\right)^2 < r_0 \tag{28}$$

which would be true for the samples that are closer to the mean μ_2^i. Because the goal is to detect outliers, however, this yields the opposite of the intended behavior, so the condition logic should be changed to

$$\left(\sigma_2^i\right)^2 > r_0 \tag{29}$$

2.2.4 Anomalous Dimensionality Penalty

The set of expressions 17 explicitly penalizes the variance and eccentricity of a microcluster by the dimensionality of the data d. This is not standard, suggested by any of the papers that introduce the TEDA metrics [2, 3], or justified by any theoretical or empirical evidence.

2.2.5 Unclear Microcluster Activation Criterion

The paper defines macroclusters as connected components of a graph, meaning the set $\mathcal{M}_j$ is strictly determined by the existence of edges between microclusters. However, the condition for the existence of an edge between two microclusters depends on the mean density of $\mathcal{M}_j$. This creates a topological contradiction: a pairwise edge cannot be evaluated using the global properties of a set that the edge itself is required to construct.

Furthermore, in an online setting, there are no macroclusters the first time the existence of an edge is checked. Because the sets $\mathcal{M}_j$ are initially empty, the term mean $\left(D_k^l\right)$ is mathematically undefined, leaving no valid mechanism to determine if the initial edges should exist.

Even assuming prior initialization, the criterion suffers from temporal recursion. When evaluating a new candidate microcluster at time k, the formulation does not specify whether the mean of $\mathcal{M}_j$ is computed over the prior topological state or the posterior state. Although the former is more intuitive, the formal notation indicates the latter.

Finally, the inequality contains a fundamental notational flaw. In the expression $D_k^l \geq \text{mean}\left(D_k^l\right)$, the index l is mathematically ill-posed. It is simultaneously used to denote a specific, isolated microcluster on the left side of the inequality and as a dummy iteration variable for the set mean on the right side.

2.3 microTEDAclus Updates in the *Mineração de Dados Aplicada a Problemas de Last-Mile na Logística* PHD Thesis

From the algorithm [6] presents, only the changes from the original microTEDAclus are described in this section.

In it, the condition for outlier detection in microclusters with two samples is changed to

$$\left(\zeta_2^i(x_2) > \frac{m^i(2)^2 + 1}{4}\right) \wedge \left(\left(\sigma_2^i\right)^2 > r_0\right) \tag{30}$$

which solves the contradiction of the r_0 usage, but retains the issue of the incorrectly used conjunction.

An explicit decay logic is introduced. Microclusters have a new parameter v_k^i, initialized as $v_k^i = 1$ when the microcluster is created, and updated as

$$v_k^i = v_{k-1}^i - \rho \tag{31}$$

when the sample x_k is an outlier to the microcluster i, where $\rho = \frac{1}{\text{decay}}$ and *decay* is a hyperparameter of the model. A microcluster is removed if $v_k^i \leq 0$.

To adequate the algorithm to the problem's data projection, the euclidean distance is replaced by the cosine similarity, which [3] states to be a valid proximity function for the TEDA metrics. The outlier condition is changed to

$$\left(x_k - \mu_k^i\right)^2 > \left(m_k^i\right)^2 \sigma_2^i \tag{32}$$

And the microcluster-update expressions are changed to

$$\begin{aligned}
S_k^i &= S_{k-1}^i + 1 \\
\mu_k^i &= \frac{S_k^i - 1}{S_k^i}\mu_{S_k^i - 1}^i + \frac{1}{S_k^i}\frac{x_k}{\|x_k\|}, \quad \|x_k\| \neq 0 \\
\left(\sigma_k^i\right)^2 &= \frac{S_k^i - 1}{S_k^i}\left(\sigma_{k-1}^i\right)^2 + \frac{1}{S_k^i - 1}\left(\frac{2\|x_k - \mu_k^i\|}{d}\right)^2 \\
\xi^i(x_k) &= \frac{1 - \mu_k^i \frac{x_k}{\|x_k\|}}{1 + k\left(\mu_k^i\right)^2}, \quad \|x_k\| \neq 0
\end{aligned} \tag{33}$$

The remainder of the algorithm, including the macrocluster formation process, is unchanged.

2.3.1 Critique

2.3.2 Issues not Addressed by the Updates

The issues in sections 2.2.2, 2.2.4 and 2.2.5 are completely ignored in the updated model, and the issue 2.2.3 is only partially solved, as the conjunction is still present in the condition for outlier detection in microclusters with two samples.

2.3.3 Import of an Incompatible Outlier Condition from Angelov's Paper

[2] derives 13 from 32 explicitly for the euclidean distance, and states that a similar condition can be derived for the cosine similarity, but does not provide any such condition. The employment of such an incompatible condition in the updated model is not justified by any theoretical or empirical evidence, and it suggests a fundamental error in the algorithm's design.

2.3.4 Usage of Euclidean Variance with Cosine Similarity

Although it updates the eccentricity formula to be compatible with the cosine similarity, the updated model retains the original variance computation 33, which is explicitly derived for the euclidean distance. There is no theoretical or empirical evidence that this variance computation is compatible with the cosine similarity, and it suggests another fundamental error in the algorithm's design.

3 Reassessment

3.1 Data

3.2 Preprocessing

3.3 Processing

3.4 Experiments

4 Proposal for an Improved Model

References

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